

Data Architecture Documentation

1. Data Architecture Overview

The project utilizes a **Relational Data Model** implemented within Microsoft Power BI. The architecture integrates multiple disparate data sources (IEA, Our World in Data, World Bank) into a unified schema to facilitate cross-functional analysis of Energy Generation, Emissions, and Economic indicators.

2. Logical Schema Design

The model follows a **Constellation Schema** approach, connecting multiple Fact Tables (Generation, Emissions, GDP) through shared relationships (Country, Year).

- **Primary Key (PK):** Country and Year
- **Relationships:** One-to-Many (1:*) relationships are established between lookup tables (Countries) and transactional tables (Data).

3. Physical Table Structure

A. Fact Tables (Transactional Data)

- **Compare between the emissions r... (Main Generation Table)**
 - *Description:* Primary dataset containing electricity generation values in TWh.
 - *Key Columns:* Country, Year, Electricity from gas, solar, wind, hydro, nuclear.
 - *Usage:* Calculate “Energy Mix” and “Renewable Share.”
- **Co2-data & owid-co2-data (Emissions Tables)**
 - *Description:* Historical carbon emission data sourced from Our World in Data.
 - *Key Columns:* co2, co2_per_capita, cement_co2, gas_co2, oil_co2.
 - *Usage:* Calculate “Grid Carbon Intensity” (Mt/TWh).
- **GDP, PPP (constant 2021...) (Economic Table)**
 - *Description:* GDP data adjusted for Purchasing Power Parity.
 - *Key Columns:* GDP, Year, Country.
 - *Usage:* Measure “Economic Energy Efficiency” (GDP per Unit of Energy).
- **Power consumption per... (Demand Table)**
 - *Description:* Electricity consumption metrics.
 - *Key Columns:* Electricity consumption per person, Units.

B. Dimension Tables (Context & Filters)

- **Countries PG (Peer Group Dimension)**
 - *Description:* Identifies countries within $\pm 10\%$ generation capacity of Egypt.
 - *Key Columns:* Country, Difference in Total, Close Countries to Egypt.
- **Population (Demographic Dimension)**
 - *Description:* Total population data for normalization.
 - *Key Columns:* Population, Year, Country.
- **Egypt PG (Focus Dimension)**
 - *Description:* Subset of data isolated for the “Egypt Overview” dashboard.
 - *Usage:* Ensures high-performance rendering of local KPIs.